

- Q.26 Explain the concept of two plate mould.
- Q.27 Explain the advantages of power assisted polishing kits.
- Q.28 Give the principle of ejection and feed system mould.
- Q.29 Describe the main principles of the injection moulding process and its parameters.
- Q.30 What are the primary parts of a compression moulding machine?
- Q.31 Describe the difference between manual polishing and power-assisted polishing techniques.
- Q.32 Discuss the importance of accurate cost estimation in moulding production planning and budgeting.
- Q.33 Explain the importance of routine maintenance in increasing the life of injection moulding machines.
- Q.34 Explain the importance of correct mould material for achieving quality moulded parts.
- Q.35 Discuss the process of specifying a mould for a specific plastic component.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain the function of transfer moulding machine. Explain its various parts.
- Q.37 Explain the various elements of mould with diagram and give their functions.
- Q.38 Discuss the importance of establishing a safety in a manufacturing facility and how it increases overall productivity.

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2nd Year / Advance Diploma in Tool and Die Making Subject:- Tool Design Theory-II (Plastic Moulds)

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which of the following cooling systems is used in injection moulding process to increase solidification rate of components made?
- Air jet cooling system
 - Water cooling system
 - Cooling with free convection
 - Cooling with fins
- Q.2 Credit cards are made of
- Polyethylene
 - Polypropylene
 - Polyvinyl chloride (PVC)
 - Polystyrene
- Q.3 How does the piston in clamping unit move?
- Hydraulic energy
 - Pneumatic energy
 - Heat energy
 - Suction
- Q.4 Which of the following is not a power operated molding machine?
- Squeeze machine
 - Sand slinger
 - Jolt-squeeze machine
 - Jot-sand machine

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- Q.5 Which of the following helps connecting runner and the mold cavity?
- a) Sprue b) Riser
c) Gate d) Pouring cup
- Q.6 Which of the following is not a part of injection moulding machine?
- a) Ejector pins b) Sprue
c) Nozzle d) Riser
- Q.7 For the functioning of the gating system, which of the following factors need not be controlled?
- a) Type of sprue b) Size of runner
c) Temperature of molten metal
d) Type of riser
- Q.8 Material commonly used for making cooling channels in injection moulds
- a) Aluminum b) Brass
c) Stainless steel d) Copper
- Q.9 The primary purpose of safety measures in a manufacturing environment
- a) Enhancement productivity
b) Reduce maintenance costs
c) Ensure worker well-being
d) Improve material quality
- Q.10 Which of the following is not a standard maintenance procedure for injection moulding machines?
- a) Lubrication of moving parts
b) Cleaning of cooling channels
c) Calibration of temperature controllers
d) Painting of mould surfaces

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Define machine time estimation for mould making.
- Q.12 What is process set-up data.
- Q.13 What is a check-list?
- Q.14 What plastic is used to laminate kitchen worktops?
- Q.15 Define trimming.
- Q.16 Define cycle time for a mould.
- Q.17 What is thermoforming?
- Q.18 The purpose of the cooling system in injection moulding is to _____ the mould and solidify the product.
- Q.19 The concept of _____ involves implementing measures to prevent accidents and injuries in the workplace.
- Q.20 The core and cavity are the primary components responsible for forming the _____ of the product.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain the main features of blow mould.
- Q.22 Explain the ejection system for compression mould with neat sketch.
- Q.23 Write short note on mould cooling system.
- Q.24 Write a short note on transfer moulding.
- Q.25 Describe blow mould and write its applications.